

Eric Ning

ningeric12345@gmail.com | [linkedin.com](https://www.linkedin.com) | github.com

EDUCATION

University of Waterloo

Software Engineering

Waterloo, ON

Sep. 2026

EXPERIENCE

UW Orbital

CubeSat Design Team

May 2026 – present

Waterloo, ON

- Used FastAPI to implement user endpoints to modify user details and sync user information to the current Keycloak service
- Implemented main table endpoints to retrieve command and telemetry data from CubeSat
- Configured Keycloak to support integration tests and created a .yaml workflow to verify its service startup for auth
- Created pytest on the functionality of login, callback, and logout API auth endpoints, and GET, PATCH, and DELETE user request endpoints

MCPT Vice President

Mackenzie Computer Programming Team

Sep 2024 – Jun 2026

North York, ON

- Created/tested 100+ competitive programming problems in C++, Java & Python, and organized them into contests
- Managed and trained contest executives, assigning them problems to create and test
- Taught full classrooms dozens of challenging competitive programming concepts including DP and Fenwick Trees

Kumon Centre Assistant

Kumon of Richmond Hill-Elgin Mills

Jun 2025 – present

Richmond Hill, ON

- Supervised/tutored 60+ students daily in both Math and Reading ranging from kindergarten to university level
- Marked students' classwork and homework, providing detailed constructive feedback for them to work on

TOPSOJ Volunteer Problem Tester

TOPS Online Judge

Dec 2023 – present

North York, ON

- Tested contest mathematics problems including math olympiad level problems (e.g., CMO level)
- Tested for dozens of math contests with up to 60+ participants and up to \$1,500+ prizes

PROJECTS

stratum | *JavaScript, Flask, Python, HTML/CSS*

- Developed a full-stack chess website, utilizing browser storage to synchronize UI states across webpages
- Implemented 20 chess modifiers and 70+ uniquely defined challenges for chess players
- Implemented chess engines running Stockfish within Flask with minimal API calls from the JavaScript frontend

minidungeon | *Python*

- Developed a basic dungeon-crawler-like game with different stages using the Python pygame module
- Created dynamic gameplay including randomized minigames and mathematical state tracking to scale difficulty
- Implemented scalable object-oriented design for entities and UI states

blahajsweeper | *Python*

- Developed a simplistic object-oriented minesweeper-like game using the Python pygame module
- Implemented structured controls to handle transitions between menus, gameplay event loops, and game conditions

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, HTML/CSS, SQL (Postgres)

Frameworks & Tools: FastAPI, Flask, Docker, Git, VS Code, Visual Studio, PyCharm, Eclipse

AWARDS AND HONOURS

USACO Gold

CCC Senior Group 2 (Top 50)

2026 CyberTitan National Finals - 1st Place

Governor General's Academic Medal - Highest Academic Average

Chess National Master (CFC)